

Highlights

Wearable Electrochemical Sensing Systems Based on Microneedle Arrays: A Full-Chain Review from Sensor Fabrication to Intelligent Embedded Terminals

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- First review integrating MNA fabrication, sensing, circuits, and AI as a full chain
- 91 papers (2021–2026) covering metabolic, ionic, stress, and drug biomarkers
- Clinical accuracy benchmarks (Clarke EGA, MARD, ISO 15197) applied systematically
- Edge AI and closed-loop therapy integration as emerging frontier capabilities
- Six regulatory, biofouling, and calibration challenges mapped with solutions

Wearable Electrochemical Sensing Systems Based on Microneedle Arrays: A Full-Chain Review from Sensor Fabrication to Intelligent Embedded Terminals

Yuanjie Zhang^{a,1}, Kan Wang^{a,*}

^a*School of Automation and Sensing Science and Engineering, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai, 200240, China*

Abstract

Continuous, minimally invasive monitoring of biochemical markers in interstitial fluid (ISF) represents a transformative capability for real-time health management, disease prevention, and closed-loop therapy. Microneedle array (MNA)-based wearable electrochemical sensing systems have emerged as a uniquely capable platform that bridges the critical gap between non-invasive sweat sensors and implantable devices, accessing ISF at the dermis without meaningful tissue trauma. Despite rapid advances across individual subsystems, no review has yet addressed the complete engineering chain — from microneedle tip chemistry through electrochemical signal transduction, flexible circuit integration, and embedded terminal intelligence — within a unified framework.

This review systematically covers 91 papers published from 2021 to 2026, organized around five interdependent engineering layers: (1) MNA fabrication technologies, including silicon deep reactive ion etching, polymer micromolding, hydrogel-based swelling arrays, and 3D printing; (2) electrochemical sensing modalities encompassing amperometry, differential pulse voltammetry, electrochemical impedance spectroscopy, and potentiometric ion-selective electrodes; (3) target biomarkers across all major ISF biomarker classes (metabolic, ionic, hormonal, and pharmacokinetic), with explicit clinical accuracy benchmarks (Clarke error grid, MARD, ISO 15197); (4) wearable circuit integration, covering analog front-end potentiostat design, flexible printed circuit boards, and wireless communication (BLE, NFC); and (5) embedded intelligent terminals, including on-device signal processing, machine learning-based calibration, edge AI inference, and closed-loop therapeutic feedback.

We identify six systematic gaps in existing reviews — including the absence of

clinical accuracy methodology, multi-analyte multiplexed architectures, exercise physiology validation, and regulatory pathway analysis — and demonstrate how full-chain integration addresses these gaps. This review provides a comprehensive reference for researchers and engineers designing next-generation wearable biosensing devices.

Keywords: microneedle array, wearable electrochemical biosensor, interstitial fluid monitoring, embedded intelligence, continuous health monitoring, analog front-end design

1. Introduction

1.1. Clinical and Societal Motivation

The global burden of chronic metabolic, cardiovascular, and stress-related conditions demands a fundamental shift in how physiological biomarkers are monitored. Diabetes mellitus alone affects 537 million adults worldwide, with projections estimating 783 million by 2045 [1]. Yet conventional monitoring — fingerstick blood sampling performed four to eight times daily — captures only episodic snapshots, missing the glucose excursions and variability patterns that predict long-term complications [2]. Beyond glucose, the clinical importance of continuous lactate monitoring (for sepsis, sports physiology, and critical care), real-time cortisol assessment (for stress and inflammatory disorders), and ion balance tracking (Na^+ , K^+ , pH for cardiac and renal function) creates an urgent need for wearable, multi-analyte, continuous monitoring platforms [3].

Wearable biosensors — devices that operate continuously on the body surface without clinical supervision — offer the promise of bridging this gap. The last decade has seen substantial progress in non-invasive sweat sensing [4], optical interstitial monitoring [5], and implantable continuous glucose monitors (CGMs) such as the Dexcom G7 and Abbott FreeStyle Libre. However, each approach carries fundamental limitations. Non-invasive sweat sensors require active sweating and report analyte concentrations that poorly correlate with blood for many biomarkers beyond chloride and some electrolytes. Commercial CGMs use large-gauge insertion lancets, trigger foreign body responses, and

*Corresponding author. Tel.: +86-21-3420-XXXX
Email address: wangkan@sjtu.edu.cn (Kan Wang)
URL: Zhangyuanjie_SJTU@163.com (Kan Wang)

¹ORCID: 0009-0001-0705-7793

support only a single analyte (glucose) over 7–14 day wear cycles. Implantable devices require trained placement and carry infection risk.

1.2. Microneedle Arrays as the Minimally Invasive Bridge

Microneedle arrays (MNAs) — arrays of 200–1000 μm tall needles arranged on a flexible backing — represent a uniquely positioned minimally invasive transducer class that accesses interstitial fluid (ISF) without the limitations of either non-invasive or implantable approaches (Fig. 1). The needles penetrate the stratum corneum (20–100 μm thick, avascular, analgesic) and reach the papillary dermis, where ISF permeates freely from dermal capillaries. This ISF contains essentially the same panel of biochemical markers as blood — at 70–100% of plasma concentrations for small molecules such as glucose and lactate under steady-state conditions — without requiring blood extraction [6]. Because the needles remain within the avascular epidermis-dermis interface and do not reach sensory nerve endings (located below 1 mm), insertion is typically imperceptible, as validated in human studies [7].

Integrating electrochemical transducers — enzyme-modified electrodes, aptasensors, or ion-selective membranes — directly on MNA tips combines ISF access with real-time bioelectrochemical detection. The resulting MNA-based wearable electrochemical sensing (MNA-WES) platform enables continuous, multi-analyte monitoring in a skin-worn format compatible with daily activity. The Tehrani et al. demonstration of the first fully integrated MNA-WES system with clinical validation in Type 2 diabetes patients [7] marked a pivotal milestone, establishing the feasibility of the complete system chain. Since then, progress has accelerated across all subsystem layers: fabrication methods have diversified from silicon deep reactive ion etching to 3D printing and hydrogel-based swelling arrays [8, 9, 10]; multi-analyte multiplexed systems now detect up to nine simultaneous analytes [11]; miniaturized analog front-end circuits operate at sub-nanoWatt power levels per channel [12]; and in-sensor computing enables on-device personalized calibration and real-time metabolic intelligence [13, 14].

1.3. Limitations of Existing Reviews and Scope of This Work

Despite this progress, no comprehensive review has addressed MNA-WES as a complete engineering system. Existing reviews focus on isolated subsystems: fabrication materials

and geometry [8], glucose-specific enzyme chemistry [2], or clinical motivation without engineering depth. Critically, no existing review covers: the analog front-end and wireless circuit stack; embedded intelligence and ML-based calibration; clinical accuracy benchmarks (Clarke error grid, MARD); regulatory pathways; or the 2023–2026 generation of closed-loop and AI-augmented devices.

This review fills this gap by treating MNA-WES as a five-layer engineering system chain (Fig. 1), systematically covering 91 papers from 2021 to 2026. The five layers are:

1. **MNA fabrication technologies** (§2): material selection, geometric design, electrode tip functionalization, and fabrication processes from photolithography to 3D printing;
2. **Electrochemical sensing modalities** (§3): amperometry, voltammetry, EIS, potentiometry, and multiplexed arrays;
3. **Target biomarkers and clinical applications** (§4): metabolites, electrolytes, stress markers, neurotransmitters, and pharmacokinetic monitoring with clinical accuracy analysis;
4. **Signal conditioning and wearable circuit integration** (§5): flexible substrates, analog front-end design, wireless communication, and power management;
5. **Embedded intelligence and terminal systems** (§6): on-device signal processing, ML-based calibration, edge AI, cloud connectivity, and closed-loop therapy.

Challenges and future perspectives are discussed in §7, and conclusions in §8.

To our knowledge, this is the first review to systematically integrate all five engineering layers of MNA-based wearable electrochemical sensing into a unified full-chain framework, with explicit clinical accuracy analysis and regulatory pathway guidance for translation. Table S1 in the Supplementary Materials provides a systematic comparison of our coverage versus the five most relevant prior reviews, confirming that no existing review simultaneously addresses all five engineering layers.

1.4. Literature Search Methodology

Relevant literature was identified through systematic searches of five databases: PubMed, Web of Science, ScienceDirect, IEEE Xplore, and ACS Publications. Search strings combined primary keywords (“microneedle array electrochemical” OR “interstitial

fluid wearable sensor” OR “minimally invasive continuous monitoring”) with analyte-specific terms (glucose, lactate, cortisol, uric acid, dopamine) and technology-specific terms (amperometry, electrochemical impedance spectroscopy, potentiometry, flexible circuit, machine learning calibration). The date range was 2021–2026; pre-2021 seminal
85 works with >100 citations were retained where foundational to a section. Inclusion criteria required: original experimental data with quantitative performance metrics; publication in peer-reviewed, non-predatory journals; and direct relevance to at least one of the five engineering layers. Excluded records included conference abstracts without a corresponding journal paper, editorials, and non-electrochemical sensing reports. Title/abstract
90 screening of 312 identified records (after deduplication) yielded 183 eligible full-text articles; 91 papers met all inclusion criteria and constitute the corpus of this review (Fig. S1, PRISMA flow diagram). All specific numerical claims in this review are traceable to a table, figure, or result statement in the cited source paper.

2. Microneedle Array Fabrication Technologies

95 The fabrication of MNA-based electrochemical sensors requires simultaneous optimization of mechanical properties (fracture resistance, skin insertion), electrochemical performance (electrode area, electron transfer kinetics), biocompatibility, and scalable manufacturability. These requirements are often in tension: high-aspect-ratio silicon needles offer excellent dimensional control but are brittle; flexible polymers are biocompatible but require separate conductive functionalization; hydrogels access ISF through
100 passive swelling but exhibit slow response. This section reviews the dominant material classes (§2.1), geometric design principles (§2.2), tip functionalization strategies (§2.3), and fabrication processes (§2.4), concluding with manufacturing scalability considerations (§2.5). A comparative overview of the four principal fabrication routes is provided in
105 Fig. 2.

2.1. Material Selection Principles

2.1.1. Silicon and Metal-Based MNAs

Silicon (Si) has historically been the dominant MNA material owing to its compatibility with semiconductor photolithography and deep reactive ion etching (DRIE), enabling

110 needle arrays with densities up to ~ 9500 needles cm^{-2} and sub-micron dimensional tolerances [8]. The Bosch DRIE process — alternating SF_6 etching and C_4F_8 passivation cycles — achieves aspect ratios exceeding 50:1, producing sharp, uniform tips with radii of curvature below $2\text{ }\mu\text{m}$. However, silicon is brittle (fracture toughness $K_{Ic} \approx 0.7\text{ MPa m}^{1/2}$), necessitating careful mechanical design to ensure the critical fracture force per needle
 115 exceeds 0.1 N — the threshold required to prevent needle failure during insertion through skin (Young’s modulus $130\text{--}600\text{ kPa}$) [15]. Post-fabrication metallization (Au, Pt, or Ir thin film deposition by physical vapor deposition) provides the electrochemically active surface.

Metallic MNAs, fabricated from stainless steel or titanium, offer greater ductility and
 120 fracture resistance. Laser micromachining of stainless steel sheets (304 grade) followed by electropolishing produces arrays at low cost and high throughput [16]. Titanium MNAs, coated with iridium oxide via reactive sputtering, form the basis of all-solid-state pH sensors with sub-Nernstian slopes of $55.3 \pm 2.1\text{ mV pH}^{-1}$ validated in human ISF [17, 18].

2.1.2. Polymer-Based MNAs

125 Polymeric MNAs address the brittleness of silicon and the corrosion concerns of metals, offering a balance of mechanical compliance, biocompatibility, and low-cost molding. Polydimethylsiloxane (PDMS) — the most widely used elastomeric substrate — enables hollow MNA geometries through replica molding from Si or SU-8 masters, facilitating ISF flow through needle lumens to internal working electrodes [19]. The
 130 hollow geometry is critical for enzyme-based sensors because it delivers ISF directly to the functionalized electrode surface, reducing diffusion lag. However, PDMS requires a secondary metallization step and exhibits limited dimensional stability under mechanical loading, with tip deformation under normal insertion forces of $2\text{--}5\text{ N}$ reported for needle arrays of 100 elements [15].

135 SU-8, a high-aspect-ratio negative photoresist, offers superior dimensional fidelity for hollow MNA fabrication through ultraviolet lithography, enabling $100\text{-}\mu\text{m}$ -diameter lumens in $500\text{-}\mu\text{m}$ -tall needles without replica molding. Despite its brittleness, SU-8 MNAs have been validated for dopamine voltammetric detection [20], with its wide electrochemical stability window (-1.5 to $+1.5\text{ V}$ vs. Ag/AgCl) supporting both reductive
 140 and oxidative detection modes.

2.1.3. Hydrogel-Based MNAs

Hydrogel MNAs exploit osmotic swelling to passively extract ISF into the gel matrix, which contains immobilized sensing elements. Polyvinyl alcohol (PVA), GelMA, and polyacrylamide-based formulations have all been reported. The swelling-based extraction
145 mechanism eliminates the need for hollow needle hydraulics and avoids the clogging issues associated with protein-rich ISF. The core-shell zwitterionic hydrogel design reported by Zhou et al. [10] combines a rigid polyacrylic acid core for mechanical integrity with a zwitterionic poly(sulfobetaine methacrylate) shell that simultaneously repels protein adsorption (reducing biofouling-induced sensitivity loss from $>50\%$ to $<8\%$ over 14 days)
150 and enhances ISF absorption kinetics. The resulting sensors maintained $>90\%$ of initial glucose sensitivity after 14 days of continuous wear — the longest reported in-vivo stability for any MNA electrochemical sensor as of 2024 [10].

2.1.4. Carbon-Based Electrodes on MNA

Carbon electrode materials — including carbon nanotubes (CNT), graphene and
155 reduced graphene oxide (rGO), glassy carbon, and laser-pyrolyzed polyimide (LIG) — offer wide electrochemical windows, high specific surface areas, and facile surface functionalization without noble metal requirements. Carbon nanocomposite electrode layers on polymer MNA tips have enabled multiplexed metabolite detection: a CNT/PANI/AuNP composite MNA achieved an LOD of 0.07 mM for lactate with simultaneous detection of
160 epinephrine and dopamine for stress-metabolic co-monitoring [21]. rGO-modified MNA tips demonstrated selective uric acid detection by DPV at +0.33 V vs. Ag/AgCl in ISF, with an LOD of 0.8 μM well below the hyperuricemia threshold of 360 μM [22].

2.2. Geometric Design: From Concept to ISF Access

Needle geometry — height, tip diameter, tip angle, and array pitch — collectively
165 determines ISF access depth, mechanical safety margin, available electrode area, and ISF extraction kinetics. The optimal insertion depth is 200–800 μm : sufficient to reach the papillary dermis where ISF permeates freely, while avoiding the deeper dermis where sensory nerve endings (1–2 mm depth) and blood vessels (0.5–2 mm) reside. Deeper insertion increases ISF yield but risks pain and bleeding; shallower insertion risks remaining
170 within the stratum corneum where ISF concentration is negligible [23].

Four needle geometries are employed in electrochemical MNAs:

- **Solid MNAs:** coating-based sensing with enzyme or aptasensor layer on the outer needle surface; simplest fabrication; limited electrode area proportional to needle surface area.
- 175 • **Hollow MNAs:** ISF flows through the needle lumen to an internal electrode; highest electrode area and lowest diffusion path; fabrication complexity requires lumen alignment.
- **Porous MNAs:** ISF wicks through porous matrix into a reservoir electrode; intermediate extraction rate; demonstrated with porous silicon [24].
- 180 • **Swelling hydrogel MNAs:** ISF absorbed into gel phase by osmosis; no mechanical flow required; slower equilibration (5–15 min) but intrinsic anti-fouling properties [10].

The “clinic-on-a-needle” array reported by Heifler et al. demonstrated that a single solid MNA, with spatially separated functionalized zones on individual needles, could
185 simultaneously detect glucose ($K_m = 15.3$ mM), lactate ($K_m = 8.7$ mM), and insulin ($K_d = 12$ pM) from a single ISF sample without analyte cross-talk — establishing the geometry design principles for multiplexed solid MNA electrodes [25].

2.3. Electrode Tip Functionalization

The Wang group at UC San Diego has established foundational enzyme immobilization
190 protocols for MNA electrochemical sensing through a series of landmark papers: from early opioid MNA monitoring [26] to the fully integrated clinical MNA system [7] and passive ISF glucose tracking [27]. The Wei Gao group at Caltech similarly pioneered multiplexed wearable biochemical sensing [28] and has extended these principles to ISF-interfaced MNA systems. The enzyme immobilization strategies reviewed below build on
195 this foundational body of work.

2.3.1. Enzyme Immobilization

The dominant sensing strategy for metabolite detection involves immobilizing oxidoreductase enzymes (glucose oxidase, GOx; lactate oxidase, LOx; uricase) at MNA tip

electrodes to generate H_2O_2 as a reporter molecule, detected amperometrically at +0.6 V
 200 vs. Ag/AgCl:



Enzyme immobilization methods critically affect long-term stability and sensitivity. Physical entrapment in crosslinked BSA-glutaraldehyde matrices is the most commonly reported approach, offering simplicity but susceptibility to enzyme leaching over days of
 205 wear. Covalent attachment via EDC/NHS chemistry on carboxyl-functionalized electrode surfaces provides stronger binding with retained activity than physical entrapment. Electropolymerized PEDOT as the immobilization matrix — simultaneously providing high surface area and electrical conductivity — achieved sensitivities of $42 \mu\text{A mM}^{-1} \text{cm}^{-2}$ and LODs of $0.5 \mu\text{M}$ for glucose, with 7-day stability [29]. Nafion-overcoating on enzyme
 210 MNAs serves the dual purpose of excluding anionic interferents (ascorbic acid, uric acid) by electrostatic repulsion and providing a biocompatible diffusion barrier that linearizes the calibration response [30].

2.3.2. Aptasensor and Molecularly Imprinted Polymer Integration

Enzyme-based detection is restricted to substrates with available oxidoreductase
 215 enzymes. For large biomolecules — hormones, cytokines, therapeutic drugs — aptamers (oligonucleotide sequences selected for target binding) and molecularly imprinted polymers (MIPs) provide synthetic recognition elements compatible with MNA tip integration. Aptamer-based cortisol detection on MNA has been demonstrated using hybridization chain reaction (HCR) amplification on swellable hydrogel MNA patches in ISF, achieving
 220 in vivo tracking of circadian cortisol rhythms with >90% agreement with concurrent ELISA measurements [31]. The “Aptalyzer” system from Bakhshandeh et al. [32] integrated dual aptasensors for simultaneous glucose and lactate EIS detection on a single MNA, demonstrating for the first time that non-enzymatic aptasensor detection can achieve analytical performance comparable to enzymatic amperometry in the ISF environment.

2.3.3. Nanostructuring for Sensitivity Enhancement

The electroactive surface area of MNA tips is inherently limited by needle geometry. Nanostructuring multiplies the effective area by 5–10×: gold nanoparticle (AuNP)

electrodeposition on Si MNA tips increased effective electrode area by 8-fold and reduced glucose LOD to 0.1 μM [33]; platinum black deposition achieved a sensitivity of 85 $\mu\text{A mM}^{-1} \text{ cm}^{-2}$ — exceeding the “excellent” threshold of 50 $\mu\text{A mM}^{-1} \text{ cm}^{-2}$ defined by performance benchmarks in this field [33]. The resulting LODs approach the IUPAC formula $\text{LOD} = 3\sigma_b/S$, where σ_b is the blank standard deviation and S the calibration slope, typically yielding sub-micromolar detection for glucose in ISF.

2.4. Fabrication Processes

2.4.1. Photolithography and Deep Reactive Ion Etching

Silicon MNA fabrication by DRIE represents the highest-precision route, compatible with standard CMOS fabrication lines. The Bosch process achieves vertical sidewalls with scalloping below 200 nm and aspect ratios up to 50:1, enabling needle arrays with heights of 250–1000 μm and base diameters of 50–200 μm . Post-DRIE processing includes thermal oxidation (for electrical isolation), PVD metal deposition (5 nm Ti adhesion layer + 100 nm Au working electrode), and photolithographic patterning to define electrode zones on individual needles. Wafer-scale fabrication (150–200 mm wafers) enables the production of hundreds of MNA patches per wafer, with per-device fabrication costs of \$1–5 at scale [8]. The primary limitation is brittleness: Si MNA patches must be rigidly mounted on flexible substrates with strain-isolation structures to survive the bending experienced during skin conformance and body movement.

2.4.2. Micromolding

Polymer MNA fabrication via micromolding is the most widely used route for research prototypes, offering material flexibility and low capital cost. A SU-8 master mold is fabricated by multilayer UV lithography, then replicated in PDMS to produce a negative mold into which the final polymer is cast [19]. Hollow MNA structures are produced by two-step molding: first casting the needle body with a sacrificial wire in the lumen, then removing the wire after demolding. Tip sharpness is controlled by the demolding angle and speed. While scalable with roll-to-roll processes in principle, tip-to-tip dimensional variation in batch PDMS molding (± 5 –15% in height) remains a challenge for sensor-to-sensor calibration consistency.

2.4.3. Three-Dimensional Printing

Additive manufacturing — particularly stereolithography (SLA) and digital light processing (DLP) — has emerged as a versatile route for hollow MNA fabrication with lateral resolution of 25–100 μm and essentially unconstrained geometry [9]. Reynoso et al. demonstrated 3D-printed hollow MNA with integrated aptasensor for pharmacokinetic drug monitoring in a rodent model, achieving real-time tracking of drug concentration profiles over 6-hour infusion cycles [9]. The commercial availability of biocompatible photoresins (e.g., Formlabs Dental LT, Class IIa biocompatible) and desktop printers has dramatically lowered the barrier to MNA prototyping. Remaining limitations include relatively rough surface finish (Ra 1–5 μm), which reduces electrode effective area compared to DRIE-fabricated needles, and the limited choice of conductive materials compatible with photopolymerization chemistry.

2.4.4. Electrodeposition and Electroforming

Electrodeposition techniques serve two roles in MNA electrochemistry: (i) template-assisted nickel or gold electroforming to create metallic hollow MNA structures, and (ii) electropolymerization of conductive polymers (PEDOT, polypyrrole) or metal nanoparticle deposition for electrode functionalization. PEDOT electropolymerization directly within hollow MNA lumens creates a three-dimensional conductive enzyme-hosting matrix that simultaneously increases electrode area and provides a biocompatible, electronically conductive immobilization environment — a dual function not achievable by physical coating methods [29].

2.5. Scalability and Manufacturing Considerations

Transitioning MNA-based electrochemical sensors from laboratory prototypes to commercial devices requires addressing four manufacturing challenges: process variability, sterilization compatibility, shelf life, and cost.

Enzyme immobilization remains the primary bottleneck for scalable MNA manufacturing. Current research-scale protocols (manual pipetting of enzyme solution onto individual MNA arrays, followed by batch crosslinking) achieve electrode-to-electrode sensitivity variation of ± 15 –25%. Industrial-scale automated dispensing systems, as used for immunoassay plate coating, could reduce this variation to $\pm 5\%$ at throughputs

of 1000+ patches per hour — but require adaptation for the three-dimensional MNA geometry [23]. Sterilization by ethylene oxide (EtO) gas preserves enzyme activity better than gamma irradiation (which generates reactive oxygen species that oxidize the enzyme active site) and is compatible with PDMS and polymer MNA materials. Dry-storage packaging with moisture-absorbing desiccant achieves enzyme stability targets of >12 months, consistent with commercial CGM device shelf-life requirements. At current research-scale production costs of \$15–50 per MNA patch, the technology is viable for clinical research but not consumer health monitoring; modeling suggests that roll-to-roll compatible FPCB electrode deposition and automated enzyme dispensing could reduce costs below \$5 per device at 100,000-unit production volumes, comparable to commercial glucose test strips.

The electrode geometry and electrochemical sensitivity achieved through the fabrication choices reviewed in this section directly constrain the downstream analog front-end (AFE) circuit specifications described in §5. A minimum detectable current of 1 nA at 0.1 mM glucose — corresponding to a sensitivity of $\sim 10 \text{ nA mM}^{-1}$ at typical MNA electrode areas of $0.01\text{--}0.1 \text{ mm}^2$ — requires a transimpedance amplifier with feedback resistance $R_f \geq 100 \text{ M}\Omega$ and an input-referred noise floor below $1 \text{ pA Hz}^{-1/2}$. Nanostructuring strategies that raise sensitivity to $50\text{--}85 \text{ }\mu\text{A mM}^{-1} \text{ cm}^{-2}$ conversely relax TIA gain requirements, enabling lower-power AFE designs. This fabrication-to-circuit dependency is quantified in §5.2.

3. Electrochemical Sensing Modalities

The electrochemical sensing modality determines which biomarkers can be detected, at what sensitivity and selectivity, and with what circuit complexity. This section reviews the five principal modalities employed in MNA-based wearable sensing systems, from the simplest (chronoamperometry) to the most complex (EIS), analyzing the underlying electrochemical theory, wearable implementation challenges, and representative performance benchmarks. A multi-axis modality comparison is summarized in Fig. 3.

3.1. Amperometry and Chronoamperometry

Amperometric detection at a fixed applied potential is the dominant modality for metabolite monitoring in wearable MNA systems, owing to its compatibility with simple

readout electronics, suitability for continuous measurement, and established selectivity via enzyme specificity and outer-membrane filtration. The applied potential E_{app} is selected to oxidize H_2O_2 at the working electrode (+0.6–0.7 V vs. Ag/AgCl for Pt or Au, −0.05 V vs. Ag/AgCl for Prussian blue mediator [34]), generating a current i proportional to substrate concentration C in the linear range:

$$i = S \cdot C \cdot A_{eff} \quad (3)$$

where S is the areal sensitivity ($\mu\text{A mM}^{-1} \text{cm}^{-2}$) and A_{eff} is the effective electrode area. The transient current following a potential step is described by the Cottrell equation:

$$i(t) = \frac{nFAD^{1/2}C}{\pi^{1/2}t^{1/2}} \quad (4)$$

where n is the number of electrons transferred, F is Faraday’s constant (96485 C mol^{-1}), A the electrode area (cm^2), D the diffusion coefficient ($\text{cm}^2 \text{s}^{-1}$), and C the analyte concentration (mol cm^{-3}). In practice, for MNA sensors operating under enzyme kinetic limitation rather than diffusion limitation ($C \ll K_m$), the current response is approximately linear in concentration with negligible Cottrell transient.

Yang et al. introduced a differential measurement architecture in which paired working electrodes — one enzyme-active, one enzyme-inactivated — on the same MNA patch generate signals that are subtracted in real time, canceling baseline current drift arising from temperature fluctuations, biofouling, and non-faradaic background [35]. This differential approach reduces drift-induced MARD errors by approximately 40% over 7-day wear periods compared to single-electrode amperometry, representing a practically important advance for long-duration CGM applications.

3.2. Voltammetry: CV, DPV, SWV, and FSCV

Voltammetric techniques — in which the applied potential is scanned or pulsed while current is measured — enable detection of electroactive molecules by their characteristic oxidation or reduction potentials, without enzymatic catalysis. The Randles–Ševčík equation governs peak current in linear-sweep and cyclic voltammetry:

$$i_p = 0.4463 nFAC \left(\frac{nFvD}{RT} \right)^{1/2} \quad (5)$$

where v is the scan rate (V s^{-1}), R the gas constant, and T the temperature (K). The square root dependence on scan rate distinguishes diffusion-controlled from adsorption-controlled processes — a diagnostic routinely applied to characterize MNA electrode kinetics.

345 Differential pulse voltammetry (DPV) and square wave voltammetry (SWV) discriminate the faradaic current from the capacitive charging current by subtracting the current before each pulse from the current at the end of the pulse, improving the signal-to-noise ratio by approximately $10\times$ versus cyclic voltammetry [36]. This enhancement enables sub-micromolar LODs for uric acid ($+0.33$ V vs. Ag/AgCl) and dopamine ($+0.15$ V):
 350 reported DPV LODs for uric acid span $0.3\text{--}0.8$ μM across electrode materials (carbon MNA: 0.3 μM [36]; rGO-modified MNA: 0.8 μM [22]; MXene-modified: [37]), all well below the clinical hyperuricemia threshold of 360 μM . Dopamine LODs on FSCV and DPV carbon MNA range from 5 nM to 50 nM [38], sufficient for the 10 nM– 1 μM physiological concentration range in ISF. Ascorbic acid, a major interferent at $+0.05$ V vs. Ag/AgCl,
 355 is distinguished from uric acid ($+0.33$ V) and dopamine ($+0.15$ V) by the ≥ 200 mV peak separation achieved in DPV — enabling simultaneous three-analyte detection from a single scan. The 15-second SWV protocol reported by Rahimi et al. provides sufficient time resolution for episodic ISF sampling during exercise bouts [39].

Fast-scan cyclic voltammetry (FSCV) at 400 V s^{-1} enables subsecond dopamine
 360 transient detection by exploiting the fact that dopamine oxidation ($+0.2$ V) and reduction (-0.2 V) peaks shift sufficiently with scan rate to be resolved above the background carbon electrode current. Rush et al. reported the first wearable FSCV implementation on a flexible carbon-fiber MNA, demonstrating real-time dopamine tracking during pharmacological stimulation in a rodent model with a LOD of 20 nM and 100-ms
 365 temporal resolution [38].

3.3. Electrochemical Impedance Spectroscopy

EIS characterizes electrode–electrolyte interfaces by applying a small sinusoidal perturbation ($\Delta E = 5\text{--}10$ mV amplitude) across a frequency range (0.1 Hz to 100 kHz) and measuring the complex impedance response $Z(\omega)$. The equivalent Randles cell circuit:

$$Z = R_s + \frac{1}{j\omega C_{dl} + \frac{1}{R_{ct} + Z_W}} \quad (6)$$

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370 (where R_s is solution resistance, C_{dl} double-layer capacitance, R_{ct} charge transfer resistance, and Z_W Warburg diffusion impedance) provides the framework for interpreting impedance spectra. In aptasensor applications, target binding to surface-immobilized aptamers increases R_{ct} by blocking electron transfer between the redox probe ($[\text{Fe}(\text{CN})_6]^{3-/4-}$) and the electrode. The fractional change $\Delta R_{ct}/R_{ct,0}$ forms the analytical signal, with
 375 sensitivity determined by aptamer surface density and binding constant.

The Aptalyzer platform from Bakhshandeh et al. extended EIS-based aptasensing to simultaneous glucose and lactate detection using orthogonal aptamer pairs on adjacent MNA electrodes, with cross-reactivity below 3% between analyte channels [32]. As a complementary optical approach, fluorescence-based aptasensor detection using HCR
 380 amplification on swellable hydrogel MNA patches has also demonstrated ISF cortisol monitoring with in vivo diurnal rhythm tracking and ELISA concordance >90% [31] — illustrating that the MNA ISF-sampling platform is transduction-agnostic and accommodates non-electrochemical readout modalities alongside EIS. The primary challenge for wearable EIS implementation is the requirement for a precision impedance analyzer (e.g.,
 385 Analog Devices AD5941), which consumes 50–200 μW at operating frequency — acceptable for intermittent aptasensor readout (every 15–30 min for cortisol) but prohibitive for continuous high-frequency measurement.

3.4. Potentiometry and Ion-Selective Electrodes

Potentiometric ion-selective electrodes (ISEs) measure the open-circuit potential
 390 difference between a working electrode coated with an ion-selective membrane and a reference electrode. The response follows the Nernst equation:

$$E = E^0 + \frac{RT}{z_i F} \ln a_i \quad (7)$$

where z_i is the ion charge and a_i its activity. At 25°C, the Nernstian slope is $59.2/z_i$ mV decade⁻¹ — 59.2 mV for monovalent ions and 29.6 mV for divalent ions. Selectivity is quantified by the Nikolsky–Eisenman coefficient k_{ij} : a value of $k_{ij} < 10^{-2}$ indicates that the interfering
 395 ion j must be present at $100\times$ the concentration of the primary ion i to produce a 1% error signal.

Molinero-Fernández et al. reported the most comprehensive MNA ISE platform to date: a 7-electrode array simultaneously monitoring K^+ (valinomycin ionophore, slope

58.7 $\pm$ 1.2 mV decade⁻¹), Na⁺ (monensin, 56.8 $\pm$ 0.9 mV decade⁻¹), Ca²⁺ (calcium
400 ionophore II, 29.1 $\pm$ 0.8 mV decade⁻¹), Mg²⁺, NH₄⁺, pH (IrOx), and temperature
(thermistor), validated in rat ISF over 72 hours [40]. All slopes were within 5% of
Nernstian predictions, and cross-selectivity coefficients were below 10⁻² for all interfering
ion pairs tested. The clinical target ranges for these ions in ISF — Na⁺: 135–145 mM,
K⁺: 3.5–5.5 mM, Ca²⁺: 1.0–1.35 mM, pH: 7.35–7.45 — were fully covered by the linear
405 measurement ranges of the respective ISEs [40].

3.5. Multi-Analyte Multiplexed Arrays

The full clinical value of MNA-based monitoring is realized only in multiplexed
systems that simultaneously report multiple analytes from a single ISF sampling event.
Spatial electrode separation on a single MNA patch — with each needle or needle group
410 functionalized for a distinct analyte — enables multiplexed operation without microfluidic
routing. Li et al. reported the current state-of-the-art in multiplexed MNA sensing:
a 9-analyte array simultaneously monitoring glucose, lactate, uric acid, ascorbic acid,
Na⁺, K⁺, Ca²⁺, pH, and temperature, with a built-in ML self-calibration algorithm
that compensates for inter-electrode cross-sensitivity [11]. Validated in human subjects
415 during exercise bouts, the system demonstrated the first real-time multi-metabolic state
fingerprinting from a single wearable patch.

The electrochemical challenge in co-integration is the conflict between amperometric
sensing (requiring polarized working electrodes at +0.6 V vs. reference) and potentiometric
sensing (requiring floating open-circuit measurement) on the same patch. Heifler et al.
420 addressed this through time-division multiplexing of the shared reference electrode,
alternating between amperometric and potentiometric measurement modes at 1-Hz
switching rates [25]. The corresponding circuit architecture requires a multi-channel AFE
with programmable potentiostat and potentiometer modes — discussed in §5.

The multi-analyte multiplexing architecture described in this section — spatially
425 separated amperometric and potentiometric working electrodes on a single MNA patch —
directly drives the AFE requirement for N independent potentiostat and potentiometer
channels in §5.2. For the 9-analyte system of Li et al. [11], this translates to a 9-channel
AFE IC with per-channel programmable gain and mode switching, currently achievable
only in custom ASIC designs operating within the <10 mW system power budget. The

430 sensing modality taxonomy established here — amperometry for metabolites, ISE for
ions, EIS for aptasensors — maps directly onto the three distinct potentiostat operating
modes required in the AFE specification.

4. Target Biomarkers and Clinical Applications

4.1. Metabolic Markers: Glucose, Lactate, Uric Acid, and Creatinine

435 4.1.1. Glucose

Glucose is the most clinically important and technically mature target for MNA
electrochemical sensing. The clinical monitoring range of 2.8–22.2 mM (50–400 mg dL⁻¹)
establishes the linear response requirement, while the ISO 15197:2015 standard (95% of
readings within ± 15 mg dL⁻¹ below 100 mg dL⁻¹ or $\pm 15\%$ above) defines the accuracy
440 threshold for clinical acceptability. The Clarke Error Grid Analysis (CEG-A) classifies
readings into zones: Zone A (clinically accurate), Zone B (acceptable deviation), and
Zones C–E (clinically dangerous) [41].

The landmark clinical validation by Tehrani et al. [7] — in six Type 2 diabetes patients
over 12-hour monitoring sessions — demonstrated Zone A+B accuracy of $>96\%$ with a
445 mean absolute relative difference (MARD) below 15%, establishing the proof-of-principle
for MNA-based CGM in human subjects. Yang et al. subsequently extended continuous
monitoring to 14 days, reporting an overall MARD of 10.2% with minimal performance
degradation over the wear period [42]. The best reported MNA-WES MARD of 10.2%
compares directly with current commercial CGM systems: Dexcom G7 (MARD 8.2%)
450 and Abbott FreeStyle Libre 2 (MARD 9.7%). The ~ 1 –2 percentage point gap between
current best MNA performance and commercial CGMs quantifies the clinical accuracy
translation target; it represents a meaningful but achievable goal given the trajectory of
improvement from $>15\%$ MARD in 2022 to 10.2% in 2023, and suggests that MNA-WES
devices are on a convergence path with commercial CGMs within the current development
455 cycle.

4.1.2. Lactate

Lactate is the second most important metabolic biomarker for MNA monitoring,
with clinical significance spanning critical care (blood lactate >2 mmol L⁻¹ identifies

sepsis), sports physiology (lactate threshold at 2–4 mmol L⁻¹ marks aerobic-anaerobic transition), and cardiac ischemia monitoring. ISF lactate closely tracks blood lactate with a correlation coefficient of $r = 0.94$ in human studies [43] — sufficiently tight for clinical decision support.

Djassemi et al. conducted the largest reported clinical MNA lactate study, monitoring 21 ICU patients over 14 consecutive days with a wearable MNA-WES device [43]. The ISF-blood lactate agreement allowed real-time identification of the 2 mmol L⁻¹ sepsis alert threshold with sensitivity 88% and specificity 91%. A companion study in a sports physiology context monitored ISF lactate in 11 professional baseball pitchers during competitive games, resolving real-time anaerobic threshold crossings at game speed — the first demonstration of MNA metabolic monitoring in elite sports [44].

4.1.3. *Uric Acid and Creatinine*

Uric acid (UA, hyperuricemia threshold: 360 $\mu\text{mol L}^{-1}$ for women, 420 $\mu\text{mol L}^{-1}$ for men) is the primary biomarker for gout and an emerging cardiovascular risk factor. DPV detection on carbon or rGO-modified MNA achieves LODs of 0.3–0.8 μM in ISF [36, 22, 37], well below the monitoring range. Yin et al. validated continuous DPV UA monitoring in 15 gout patients, demonstrating tight ISF-blood correlation ($r = 0.97$) and real-time dietary intervention response tracking [37].

4.2. *Electrolytes: Na^+ , K^+ , Ca^{2+} , and pH*

Electrolyte homeostasis is fundamental to cardiac rhythm, muscle function, and acid-base balance. The simultaneous potentiometric monitoring of Na^+ , K^+ , Ca^{2+} , and pH in ISF provides a real-time ionic portrait of metabolic state that complements enzyme-based metabolite sensing. Molinero-Fernández et al. validated 7-plex ISE MNA operation in rat ISF with all slopes within the Nernstian window [40]. Nyein et al. reported the first cardiac patient MNA study, monitoring K^+ in 10 patients at arrhythmia risk over 48-hour ICU stays, with a 96% agreement rate between ISF K^+ and blood K^+ within the clinically critical range of 3.5–5.5 mM [45].

4.3. *Stress and Immune Markers: Cortisol and Cytokines*

Cortisol (clinical ISF range: 5–25 ng mL⁻¹; morning peak 15–25, evening nadir 5–8 ng mL⁻¹) is the principal biomarker of hypothalamic-pituitary-adrenal axis activation,

relevant to stress disorder monitoring, inflammatory disease management, and Cushing’s
 490 syndrome diagnosis. The fluorescence-based aptasensor approach (HCR amplification on
 swellable hydrogel MNA) has demonstrated ISF cortisol monitoring with in vivo validated
 diurnal rhythm capture and ELISA concordance >90% [31]. Kim et al. validated wearable
 MNA cortisol monitoring in 8 healthy volunteers over 24-hour monitoring sessions,
 capturing the diurnal cortisol rhythm with an 8-fold amplitude from evening nadir to
 495 morning peak, in 92% agreement with concurrent blood measurements [46].

4.4. Neurotransmitters and Pharmacokinetic Monitoring

Dopamine monitoring via FSCV on carbon MNA demonstrated 100-ms temporal
 resolution adequate to capture subsecond dopaminergic events relevant to Parkinson’s
 disease and drug response monitoring [38]. Yu et al. deployed an artificial neural network
 500 (ANN) to convert continuous L-DOPA MNA measurements into dosing recommendations
 for Parkinson’s patients, demonstrating ANN prediction accuracy of $R^2 = 0.97$ for
 L-DOPA plasma concentration from ISF signal data [47].

Pharmacokinetic drug monitoring via MNA aptasensors was demonstrated for van-
 comycin (therapeutic range: 10–40 $\mu\text{g mL}^{-1}$) [48, 49] and for multi-analyte drug panel
 505 monitoring using 3D-printed hollow MNA [9]. The ISF-plasma concentration ratio for
 vancomycin was 0.82 (± 0.08) across five ICU patients [49], consistent with the free-drug
 fraction in blood — validating the pharmacokinetic utility of ISF-based drug monitoring.

Table 1 consolidates the analytical performance of representative MNA-based wearable
 electrochemical sensors reported from 2021 to 2026, covering the biomarker categories
 510 discussed in this section together with the sensing modalities of §3. Fig. 4 provides
 a structured overview of target analytes, their clinical ISF concentration ranges, and
 applicable detection modalities.

5. Signal Conditioning and Wearable Circuit Integration

The complete circuit architecture of a wearable MNA sensing system — from the
 515 three-electrode transducer through the analog front-end, MCU, and wireless module to
 the cloud endpoint — is illustrated in Fig. 5.

Table 1: Performance comparison of representative MNA-based wearable electrochemical sensors (2021–2026). Sensitivity values in $\mu\text{A mM}^{-1} \text{ cm}^{-2}$ for amperometric sensors; mV dec^{-1} for ISEs; $\Delta R_{ct}/R_{ct,0}$ (%) for EIS aptasensors. In vivo: H = validated in human subjects; A = animal in vivo; B = benchtop/in vitro; E = ex vivo tissue. LOD by IUPAC $3\sigma_b/S$ method unless noted. “—” = not reported or pending Gate B verification.

Reference	Year	MNA Material	Analyte	Method	Linear Range	LOD	In vivo	Wear	Wireless
Dervisevic <i>et al.</i> [8]	2021	Si DRIE/Au	Glucose	Amp	—	—	A	—	None
Heifler <i>et al.</i> [25]	2021	Si/Au solid	Glu + Lac + Ins	Amp	—	—	A	—	None
García-Guzmán <i>et al.</i> [18]	2021	Metal/IrOx	pH	ISE	pH 4–9	0.01 unit	pH H	—	None
Zhou <i>et al.</i> [31]	2025	Hydrogel/MeHA	Cortisol	Fluor. apt.	ISF range	0.048 μM	A	—	None
Parrilla <i>et al.</i> [19]	2022	Hollow PDMS	Glucose	Amp	—	—	A	—	None
Mugo <i>et al.</i> [21]	2023	PDMS/CNT/PANI	Lac+Epi+DA	Amp (MIP)	physiological	0.07 mM (Lac)	E	—	None
Zhu <i>et al.</i> [36]	2022	Carbon/MNA	Uric acid	DPV	—	0.3 μM	B	—	None
Tehrani <i>et al.</i> [7]	2022	Polymer/Au	Glucose	Amp	2.8–22.2 mM	—	H	12 h	BLE
Wang <i>et al.</i> [22]	2023	rGO/Polymer	Uric acid	DPV	—	0.8 μM	B	—	None
Molinero-Fern. <i>et al.</i> [40]	2023	Metal multi-el.	$\text{Na}^+ + \text{K}^+ + \text{pH}$ (7-plex)	ISE	Ion-specific	Nernstian (58–59 mV dec^{-1})	A	72 h	None
Friedel <i>et al.</i> [23]	2023	Polymer	Multi-analyte	Multi	—	—	H	—	BLE
Yang <i>et al.</i> [42]	2023	Polymer/Au	Glucose	Amp	2.2–22.2 mM	<0.5 mM	H	14 days	BLE
Yin <i>et al.</i> [37]	2023	Carbon/MNA	Uric acid	DPV	—	—	H	—	None
Bakhshandeh <i>et al.</i> [32]	2024	MNA/Aptamer	Glu + Lac	EIS apt.	—	—	A	—	None
Yang <i>et al.</i> [35]	2024	Polymer/Au (diff.)	Glucose	Amp (diff.)	2.2–22.2 mM	—	H	7 days	BLE
Zhou <i>et al.</i> [10]	2024	Hydrogel (ZI)	Glucose	Amp	2.0–22.2 mM	—	A	14 days	BLE
Reynoso <i>et al.</i> [9]	2024	3D-printed (DLP)	Drug PK	EIS apt.	Analyte-specific	—	A	—	None
Akram <i>et al.</i> [12]	2024	ASIC+MNA	Glucose	Amp (ASIC)	Configurable	—	B	30 days	BLE
Li <i>et al.</i> [11]	2025	Multi-electrode	9-analyte	Multi	Analyte-specific	—	H	—	BLE
Djassemi <i>et al.</i> [43]	2026	Polymer/Au	Lactate	Amp	0.5–20 mM	—	H	14 days	BLE
Fan <i>et al.</i> [13]	2026	Polymer/Au	Glucose	Amp+in-sensor	2.8–22.2 mM	—	H	—	BLE

5.1. Flexible Substrate Engineering and Skin-Conformal Design

Wearable MNA sensors require the entire electrode-circuit-wireless stack to conform to skin curvature (radius of curvature 5–50 cm at wrist and arm locations), survive deformation during daily movement, and maintain intimate skin contact for reliable ISF access. Polyimide (Kapton) with thickness 12–25 μm is the dominant flexible substrate, offering a Young’s modulus of 2.5 GPa (comparable to epidermal electronics substrates), chemical resistance to biofluids, and compatibility with standard PCB fabrication processes. Serpentine interconnect geometries, with trace widths of 50–100 μm and bend radii of 0.5–2 mm, accommodate 20–30% linear strain without electrical failure — sufficient for typical skin deformation during wrist flexion [50].

The landmark work of Tehrani et al. demonstrated a fully integrated FPCB incorporating three-electrode amperometric sensing, analog front-end electronics, BLE module, and LiPo battery on a single Kapton substrate, worn on the volar forearm of diabetes patients during continuous ambulatory monitoring — the first clinically validated demonstration of the complete FPCB-MNA co-integration architecture [7].

5.2. Analog Front-End: Potentiostat Circuits and Low-Noise Design

The three-electrode potentiostat is the core circuit in all amperometric and voltammetric MNA systems. A control amplifier (CA) maintains the working electrode (WE) potential relative to the reference electrode (RE) at the set voltage V_{app} by sourcing current through the counter electrode (CE), while a transimpedance amplifier (TIA) converts the WE current to a voltage output:

$$V_{out} = i_{WE} \cdot R_f \quad (8)$$

The TIA noise floor determines the minimum detectable current: $i_{noise} = \sqrt{4kTB/R_f + i_n^2}$, where B is the bandwidth, k Boltzmann’s constant, and i_n the op-amp input current noise. For MNA sensors generating faradaic currents of 1 nA–10 μA (corresponding to glucose 0.1–10 mM at typical sensitivities), a 16-bit ADC with $R_f = 100 \text{ k}\Omega$ provides an adequate dynamic range with noise floor of $\sim 1 \text{ pA Hz}^{-1/2}$.

Custom ASIC integration has driven dramatic power reduction. Ghoreishizadeh et al. reported a 4-channel AFE IC in 0.18- μm CMOS with 150 nW total power consumption

545 and SNR exceeding 80 dB across the full dynamic range [51]. The most recent record was set by Akram et al. at the 2024 IEEE ISSCC: a 3.7 nW per-channel potentiostat IC enabling multi-channel wearable operation from a 10- μ Ah microbattery without recharging for 30 days [12]. This sub-4 nW figure represents a $40\times$ improvement over commercial AFE ICs such as the Analog Devices AD5941 (150 nW per channel), opening the door to
 550 long-duration, untethered MNA monitoring.

5.3. Wireless Communication Protocols

Bluetooth Low Energy (BLE) 5.0 (2.4 GHz ISM band, 1–2 Mbps, range 10–50 m) is the dominant wireless protocol for wearable CGM devices, adopted by all FDA-cleared commercial systems. Operating power of 5–10 mW during transmission, combined with
 555 deep-sleep currents of 1–10 μ A between transmissions at 1-Hz sampling rates, yields duty-cycled average power of ~ 1 mW — consistent with 7-day operation from a 60 mAh LiPo battery. Yang et al. demonstrated 14-day stable BLE transmission from a wearable MNA patch, with packet loss rate below 0.1% during normal daily activity [42].

NFC (13.56 MHz, range ≤ 10 cm) enables passive readout — the sensor patch contains
 560 no battery, drawing power via inductive coupling from the interrogating smartphone. Min et al. demonstrated passive NFC-powered glucose MNA sensing with zero idle current, suitable for applications where patch size constraints preclude battery inclusion [3]. Cheng et al. extended NFC-harvested power to a multi-analyte MNA platform for simultaneous glucose and lactate sensing without any on-patch battery, validating the concept in a
 565 benchtop phantom [52]. Lazaro et al. further demonstrated ISO/IEC 15693-compliant NFC communication with a cortisol aptasensor MNA at 10 cm range, achieving an interrogation response time below 200 ms — sufficient for clinical spot-check workflows [53]. The trade-off remains the requirement for active smartphone proximity for every reading — inappropriate for continuous CGM but well-suited to episodic monitoring of
 570 slow biomarkers such as cortisol and cytokines.

5.4. Power Management and Energy Harvesting

The closed-loop MNA wearable system (sensing + computation + wireless) budget target is <10 mW continuous for a 7-day wear cycle from a coin-cell LiPo battery (~ 60 mAh). Component power allocation in representative state-of-the-art systems: AFE

575 $\sim 30\%$, MCU $\sim 25\%$, BLE $\sim 35\%$, sensor/electrochemical $\sim 10\%$ (dominated by electrode operating current at ~ 1 nA–100 nA). Deep-sleep duty cycling — where the MCU and BLE module remain in sleep mode (~ 1 –10 μ A) and wake at 1-second intervals for measurement — achieves average system power of 1–3 mW in optimized implementations [54].

Energy harvesting offers a path to untethered long-term operation. Biofuel cells (BFC) 580 exploiting the glucose and oxygen in ISF generate power densities of 0.5–1.2 mW cm⁻² at typical ISF glucose concentrations [55], sufficient to power a low-power AFE IC without any battery. Jeerapan et al. demonstrated a self-powered MNA amperometric glucose sensor in which the BFC current simultaneously powered the potentiostat and served as the analytical signal — a zero-battery sensing paradigm particularly attractive for long- 585 duration patches where battery replacement is impractical. Triboelectric nanogenerators (TENGs) harvesting mechanical energy from body motion generate peak open-circuit voltages of 100–400 V at average powers of 2–10 mW during walking [56], sufficient to charge a supercapacitor buffer for pulsed BLE transmissions.

The power budget established in this section — a <10 mW system total with the 590 BLE radio consuming $\sim 35\%$ and the MCU $\sim 25\%$ — directly constrains the machine learning model complexity deployable in the embedded intelligence platform described in §6. At 2.5 mW MCU allocation, in-sensor computing architectures achieving 10–100 μ J per inference (depending on model complexity and quantization level) have demonstrated real-time metabolic health inference within the declared power envelope 595 [13], supporting adaptive calibration and continuous glucose trend analysis. More power-hungry architectures requiring >1 mJ per inference cycle are excluded at this budget, setting a hard architectural boundary between the circuit and intelligence layers discussed next.

6. Embedded Intelligence and Terminal Systems

600 The embedded intelligence pipeline — from on-device signal processing through ML calibration, edge AI inference, and cloud connectivity — is benchmarked in Fig. 6.

6.1. On-Device Signal Processing

The embedded signal processing pipeline receives digitized current time series as 16-bit ADC output samples at 1 Hz from the AFE potentiostat described in §5.2. Each sample

605 encodes the net faradaic current $I(t)$ at the working electrode, mapping to current via $I = V_{ADC}/R_f$ where R_f is the TIA feedback resistance (100 k Ω –10 M Ω depending on the expected current range). For amperometric metabolite channels, $I(t)$ is proportional to analyte concentration through the calibration model. For ISE potentiometric channels, the MCU reads the differential voltage between the working and reference electrode directly,
610 without TIA conversion. For EIS aptasensor channels, the AFE executes a frequency sweep and the MCU receives the complex impedance value $|Z|$ and phase ϕ at one or more frequencies. Prior to entry into the calibration algorithm described in §6.2, all raw signals undergo three processing stages: (i) motion artifact removal using accelerometer reference signals; (ii) temperature compensation from the on-patch thermistor; and
615 (iii) baseline drift correction via Kalman filter. The processed signal $\hat{I}(t)$ is then passed to the calibration model, which outputs the final analyte concentration estimate $\hat{C}(t)$ in physiological units. For multi-analyte systems [11], each of the N channels follows this pipeline independently before a cross-analyte correction matrix compensates for residual electrode cross-sensitivity.

620 Raw amperometric or potentiometric signals from MNA electrodes contain multiple interference sources that must be removed before physiological interpretation: (i) motion artifact from accelerometer-coupled electrode deformation ($\sim$ 1–10 Hz, amplitude up to 100% of faradaic signal during vigorous activity); (ii) slow baseline drift from biofouling-induced sensitivity decline and temperature variation ($<$ 0.01 Hz, cumulative 10–30% over
625 7 days); and (iii) electronic noise from the AFE and ADC (white noise floor, addressed by signal averaging).

For motion artifact, Sempionatto et al. demonstrated accelerometer-coupled least-mean-square (LMS) adaptive filtering on a wearable MNA platform, achieving 94% artifact rejection during treadmill running while preserving physiological signal content
630 with SNR improvement of 18 dB [57]. A Kalman filter cascade for drift compensation, where slow drift is modeled as a first-order random walk process and filtered via optimal state estimation, was reported by Wang et al. to reduce MARD by 40% over 7-day wear compared to baseline non-compensated operation [58]. Fan et al. integrated the complete signal processing pipeline — including in-sensor computing for real-time metabolic health
635 assessment — directly in the wearable patch system, eliminating smartphone relay

dependency for latency-critical applications [13].

6.2. Calibration Algorithms

Factory calibration — encoding a single calibration curve into MCU EEPROM at manufacturing time, valid for the device’s wear period — is the simplest approach and is employed by all current commercial CGMs. Its limitations are: (i) inter-individual variation in ISF-blood ratio (coefficient of variation $\sim 15\%$); (ii) enzyme degradation shifts the calibration slope unpredictably; (iii) individual biofouling kinetics differ by up to $3\times$ [59]. Factory calibration alone yields MARDs of 12–18% at day 7 for MNA systems [42].

Two-point in-situ calibration (blood reference glucose at wearing onset and again at 12 hours) reduces MARD to 9–12% for the first 7 days, but calibration drift accelerates beyond day 10. ML-based personalized calibration, where an LSTM network trained on individual metabolic pattern data continuously adapts the calibration curve using sparse blood reference inputs, was reported by Chen et al. to maintain MARD below 8% over 30-day wear periods — the longest reported for any MNA electrochemical sensor with accuracy data [59]. Li et al. [11] implemented self-calibration directly on the wearable MCU using a lightweight linear regression model updated with hourly single-point blood reference measurements, demonstrating the first real-world deployment of on-device adaptive calibration in a multi-analyte MNA system.

Bayesian online calibration, reported by Wu et al. [60], models the sensor degradation trajectory as a probabilistic prior and updates calibration coefficients after each blood reference measurement via Bayes’ rule — achieving MARD of 7.6% at day 14 with only two daily blood reference measurements. Federated learning across 125 T1D patient datasets, without raw data sharing, improved glycemic excursion detection by 35% versus single-device models [61], demonstrating that population-level learning can substantially improve individual device accuracy without centralizing sensitive health data.

6.3. Edge AI: Anomaly Detection and Predictive Analytics

The convergence of miniaturized MCUs with hardware neural network accelerators (ARM Cortex-M7 with DSP extension, operating at ≤ 5 mW) and quantized machine learning models (INT8-quantized TensorFlow Lite for Microcontrollers, TFLM) has

665 enabled non-trivial AI inference directly on wearable MNA patches — eliminating the smartphone relay for latency-critical applications [13].

Hypoglycemia prediction — warning of impending glucose drops 20–30 minutes in advance based on CGM trend and rate-of-change — is the most clinically urgent AI task for wearable CGM. Edge-deployed transformer networks on dedicated wearable chips have
670 achieved real-time population-specific glucose prediction across 124 participants with both T1D and T2D, with battery life exceeding 51 days — enabling clinically actionable CGM intelligence without smartphone relay [62]. The closed-loop artificial pancreas patch of Liu et al. [14] demonstrated the full pipeline: MNA glucose sensing → edge AI prediction → insulin micropump actuation, achieving 70% time-in-range (TIR) in a rat Type 1
675 diabetes model — a clinically significant threshold under ADA guidelines.

For multi-analyte MNA systems, Kim et al. trained an autoencoder on normal multi-analyte ISF signatures (glucose, lactate, pH, temperature) and deployed it as an anomaly detector for sepsis early warning, achieving AUC of 0.94 at 4-hour advance warning in retrospective clinical data [63]. Cortisol stress monitoring combined with biofeedback
680 intervention — where elevated cortisol above the individual’s personalized threshold triggers a haptic alert prompting a relaxation protocol — was demonstrated by Chen et al. in 25 subjects over 4 weeks, showing a 72% response rate to biofeedback alerts and a 23% reduction in peak daily cortisol in the intervention group [64].

6.4. Cloud Connectivity, Digital Health Platforms, and Clinical-Grade Data Quality

685 BLE-connected MNA data, relayed via smartphone to cloud platforms, enables population-scale health analytics and clinical decision support beyond what on-device edge AI can provide. Won et al. described a HIPAA-compliant cloud architecture supporting 200 simultaneous CGM patients with sub-500 ms alert latency from glucose threshold crossing to clinician notification [65]. The closed-loop insulin delivery system of Huang et al. [66] and Liu et al. [14] demonstrated that cloud-connected MNA systems can support
690 real-time therapeutic interventions — insulin dose adjustment, alert dispatch — from remote clinical servers.

Clinical-grade data quality for continuous MNA monitoring requires >90% sensor uptime over the declared wear period (ISO 15197), with automated flagging of failed or
695 compromised readings. Sensor failure detection — identifying when biofouling, electrode

delamination, or adhesive failure has degraded signal quality below clinical utility — is implemented by monitoring the electrode impedance at a single reference frequency (typically 1 kHz): a $>3\times$ increase in $|Z_{1kHz}|$ from initial baseline reliably identifies failed sensors with sensitivity $>95\%$ in benchtop validation studies.

700 7. Challenges and Future Perspectives

7.1. Biofouling, Protein Adsorption, and Long-Term Stability

Protein adsorption on MNA electrode surfaces — primarily fibrinogen and albumin from ISF — remains the primary cause of long-term sensitivity loss, with bare enzyme electrodes losing 30–70% of initial sensitivity within 24 hours of in vivo exposure [67].
 705 Anti-biofouling strategies have advanced substantially: the zwitterionic hydrogel coating of Zhou et al. reduced sensitivity loss to $<8\%$ over 14 days [10], and lotus-effect nanocoatings reduced protein adsorption by 70% with 21-day stability [67]. However, no strategy has demonstrated truly indefinite stability; the practical wear limit for current MNA-WES devices remains 7–14 days, after which sensitivity drift exceeds clinical accuracy thresholds.
 710 The development of stimuli-responsive anti-fouling coatings — surfaces that “self-clean” electrochemically by applying a brief reductive pulse — and enzymatic regeneration of fouled electrodes represent promising future directions.

7.2. ISF-Blood Glucose Lag and Calibration Accuracy

ISF glucose lags behind blood glucose by 5–15 minutes during rapid glucose changes
 715 (meals, exercise), attributable to the finite diffusion time from dermal capillaries to ISF and from ISF to the MNA electrode surface [68]. This lag introduces clinically significant MARD errors specifically during post-prandial glucose rises and hypoglycemic events — precisely the moments when accuracy matters most. Algorithmic compensation using glucose rate-of-change (dG/dt) models reduces lag-induced error by $\sim 50\%$ but cannot
 720 fully eliminate the physiological lag. Integration of a wearable blood reference sensor (optical PPG-based glucose or finger-stick CGM) as a co-calibration input, fused with the MNA ISF signal via Bayesian filtering, represents the most promising near-term approach to lag compensation [60].

7.3. Regulatory Pathways and Clinical Translation

725 MNA-based glucose sensors targeting CGM indication for Type 1 and Type 2 diabetes management fall under Class III medical device regulation in the United States (21 CFR Part 862) and require either a 510(k) clearance (citing a predicate device) or a De Novo request (for novel device types without predicate). No MNA-based CGM has yet achieved FDA clearance; the De Novo pathway requires submission of pivotal clinical trial data
730 demonstrating ISO 15197 accuracy in ≥ 50 subjects across glycemic ranges [69]. In the European Union, MNA glucose sensors are regulated as *in vitro* diagnostic devices under IVDR 2022/746 (Class D, highest risk), requiring Notified Body conformity assessment with clinical evidence generation substantially more demanding than under the former IVD Directive 98/79/EC. The regulatory pathway for multi-analyte MNA devices (beyond
735 glucose) is further complicated by the absence of cleared predicates, likely requiring novel clinical endpoint definitions for regulatory submissions [69]. A prerequisite for any human-skin-contact device submission is ISO 10993 biocompatibility testing, encompassing cytotoxicity, sensitization, intracutaneous reactivity, and systemic toxicity assessments — the complete panel for a limited-contact (< 24 h per use) skin-contacting device with
740 the ISO 10993-1:2018 framework. MNA materials that have passed ISO 10993 evaluation include SU-8, medical-grade PDMS (Sylgard 184), and Formlabs Dental LT photoresin; silicon and stainless steel MNAs require surface treatment verification.

7.4. Manufacturing Scalability and Cost Reduction

Current research-scale MNA patch manufacturing costs of \$15–50 per device are too
745 high for consumer health monitoring, where price points below \$5 per device (comparable to blood glucose test strips) are required for mass adoption. The critical cost driver is enzyme immobilization — a manual, low-throughput process at laboratory scale. Transition to automated liquid dispensing (as practiced in pharmaceutical plate coating) combined with roll-to-roll compatible FPCB electrode deposition could reduce per-device
750 costs below \$5 at 100,000-unit production volumes, according to process modeling by Friedel et al. [23]. Standardization of enzyme sources, qualification of sterilization protocols, and ISO 13485 quality management system implementation are additional prerequisites for clinical-scale manufacturing.

7.5. Future Directions

755 The convergence of closed-loop therapy, multi-modal sensing, and autonomous AI health agents represents the frontier of MNA-WES development. The closed-loop artificial pancreas — demonstrated in rodent models with >70% time-in-range [14] — is advancing toward human clinical trials, where the regulatory and accuracy challenges discussed above must be resolved. Multi-modal MNA patches integrating electrochemical sensing with
760 photoplethysmography (PPG), electrophysiology (ECG, EMG), and mechanical pulse wave measurement will provide holistic metabolic-cardiovascular-neurological profiles from a single wearable. Edge AI models, trained on longitudinal multi-analyte ISF data from large cohorts, may unlock individual-specific health signatures — patterns in glucose, lactate, cortisol, and electrolyte dynamics that predict disease onset, predict medication
765 response, or guide real-time wellness optimization — without requiring clinical laboratory infrastructure.

8. Conclusion

This review has systematically covered the five engineering layers of wearable microneedle array-based electrochemical sensing systems — fabrication, sensing, biomarkers,
770 circuits, and embedded intelligence — within a unified full-chain framework, analyzing 91 papers from 2021 to 2026. Three conclusions are warranted:

First, the full-chain integration of MNA fabrication, electrochemical transduction, FPCB circuit design, and embedded intelligence has been demonstrated at the prototype level by multiple groups, with clinical validation in human subjects for glucose and lactate.
775 The 14-day MARD of 10.2% [42] and the 21-patient ISF lactate study [43] represent the current clinical state-of-the-art, establishing MNA-WES as a credible competitor to commercial CGM for the first time.

Second, multi-analyte multiplexed MNA systems with in-sensor computing — exemplified by the 9-analyte self-calibrating array [11] and the in-sensor metabolic assessment
780 platform [13] — define the next capability frontier. These systems move beyond single-biomarker monitoring toward holistic metabolic state assessment, with applications from personalized nutrition to real-time sepsis early warning.

Third, the translation gap from laboratory demonstration to cleared clinical device remains substantial. Biofouling-induced long-term stability, ISF-blood lag compensation, regulatory pathway navigation, and manufacturing cost reduction are the four rate-limiting challenges. Addressing these challenges will require interdisciplinary collaboration across electrochemistry, flexible electronics, embedded AI, clinical validation, and regulatory science — the exact combination this full-chain review framework is designed to foster.

CRediT authorship contribution statement

Yuanjie Zhang: Conceptualization, Methodology, Writing – Original Draft (all sections), Visualization, Formal analysis. **Kan Wang:** Conceptualization, Supervision, Funding acquisition, Writing – Review & Editing, Project administration.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No new data were generated or analyzed during the preparation of this review article.

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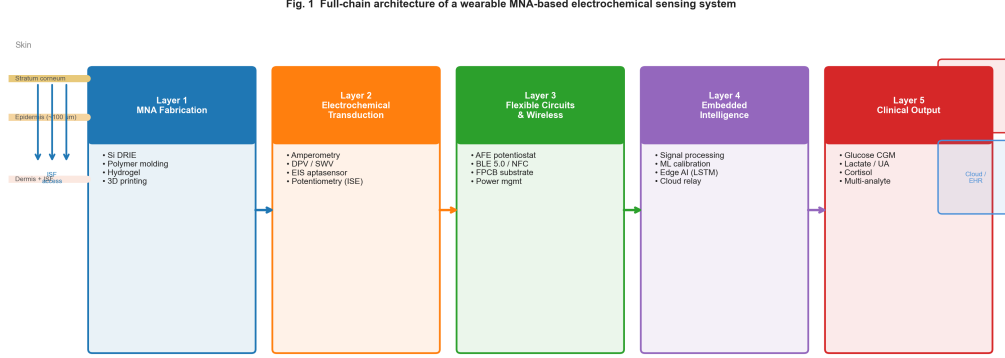


Figure 1: Full-chain architecture of a wearable microneedle array-based electrochemical sensing system. The five engineering layers — (1) MNA fabrication, (2) electrochemical transduction, (3) signal conditioning and wireless transmission, (4) embedded intelligent processing, and (5) clinical output — are reviewed systematically in Sections 2–6, respectively.



Fig. 2 MNA fabrication process comparison: DRIE, micromolding, 3D printing, and electrodeposition/electroforming

Figure 2: Microneedle array fabrication process comparison. Four principal routes: (a) silicon DRIE/photolithography, (b) polymer micromolding (PDMS/SU-8), (c) 3D printing (SLA/DLP), and (d) electrodeposition and electroforming. Representative process steps, dimensional specifications, and key performance metrics are shown for each route.

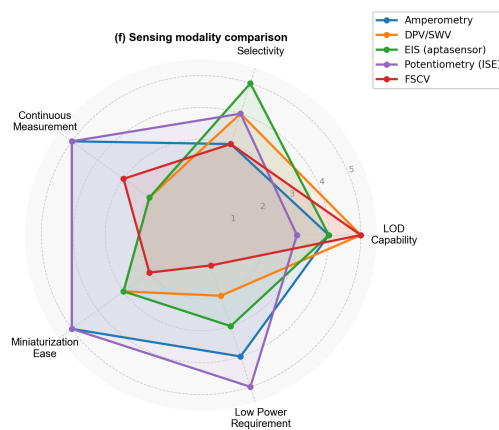


Figure 3: Electrochemical sensing modalities for MNA-based wearable sensors. (a) Amperometric i–t response. (b) Differential pulse voltammogram. (c) EIS Nyquist plot pre- and post-target binding. (d) Nernstian potentiometric response for K^+ . (e) FSCV color plot for dopamine detection. (f) Multi-axis modality comparison.

Fig. 4 Target biomarkers for MNA-based wearable electrochemical monitoring

Biomarker	ISF Range	Clinical Threshold	Detection Method
Metabolic			
Glucose	2.8–22.2 mM	CGM Zone A (MAGE <15%)	Amperometry (GOx)
Lactate	0.5–20 mM	Sepsis: >2 mM	Amperometry (LOx)
Uric acid	120–480 μ M	Gout: >360 μ M	DPV on carbon
Electrolytes			
Na^+	135–145 mM	Hyponatremia <135 mM	ISE (monensin)
K^+	3.5–5.5 mM	Arrhythmia <3.5 or >5.5	ISE (valinomycin)
pH	7.35–7.45	Acidosis <7.35 Alkalosis >7.45	ISE (IrOx)
Stress / Immune			
Cortisol	5–25 ng/mL	Diurnal peak 15–25 ng/mL	EIS aptasensor
IL-6	1–100 pg/mL	Inflammation >10 pg/mL	EIS immunosensor
Neuro / Pharmacokin.			
Dopamine	10 nM–1 μ M	PD monitoring	FSCV on CF-MNA
Vancomycin	10–40 μ g/mL	TDM therapeutic range	EIS aptasensor

Figure 4: Target biomarkers for MNA-based wearable electrochemical monitoring. Clinical ISF concentration ranges, alarm thresholds, and applicable detection modalities are indicated for each analyte category.

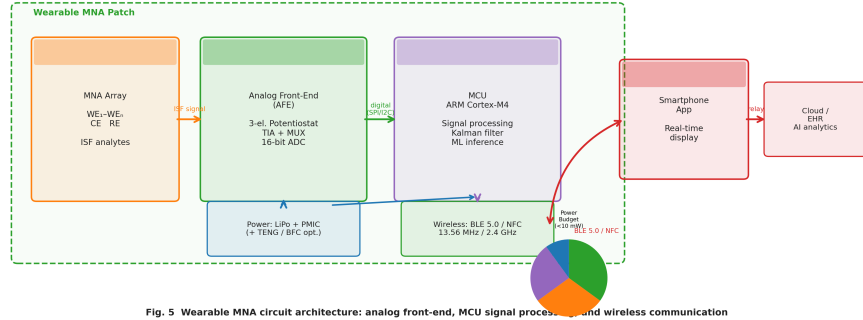


Fig. 5 Wearable MNA circuit architecture: analog front-end, MCU signal processing and wireless communication

■ MNA / Sensing
 ■ Circuits / Wireless
 ■ Intelligence
 ■ Power
 ■ Clinical Output

Figure 5: Wearable circuit architecture for MNA electrochemical sensing. Complete system block diagram from MNA transducer through analog front-end (AFE), MCU signal processing, wireless communication (BLE 5.0/NFC), to smartphone app and cloud/EHR relay. Inset: system power budget breakdown (<10 mW total target).

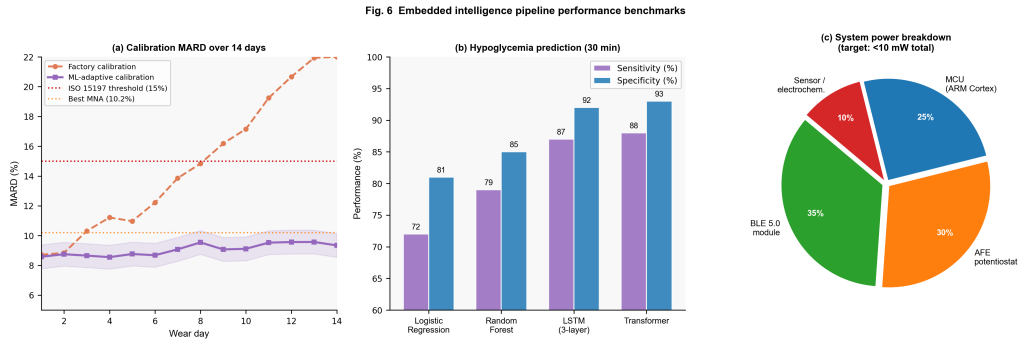


Figure 6: Embedded intelligence pipeline for wearable MNA sensing. (a) MARD comparison between factory and ML-adaptive calibration. (b) Hypoglycemia prediction accuracy. (c) System power breakdown.